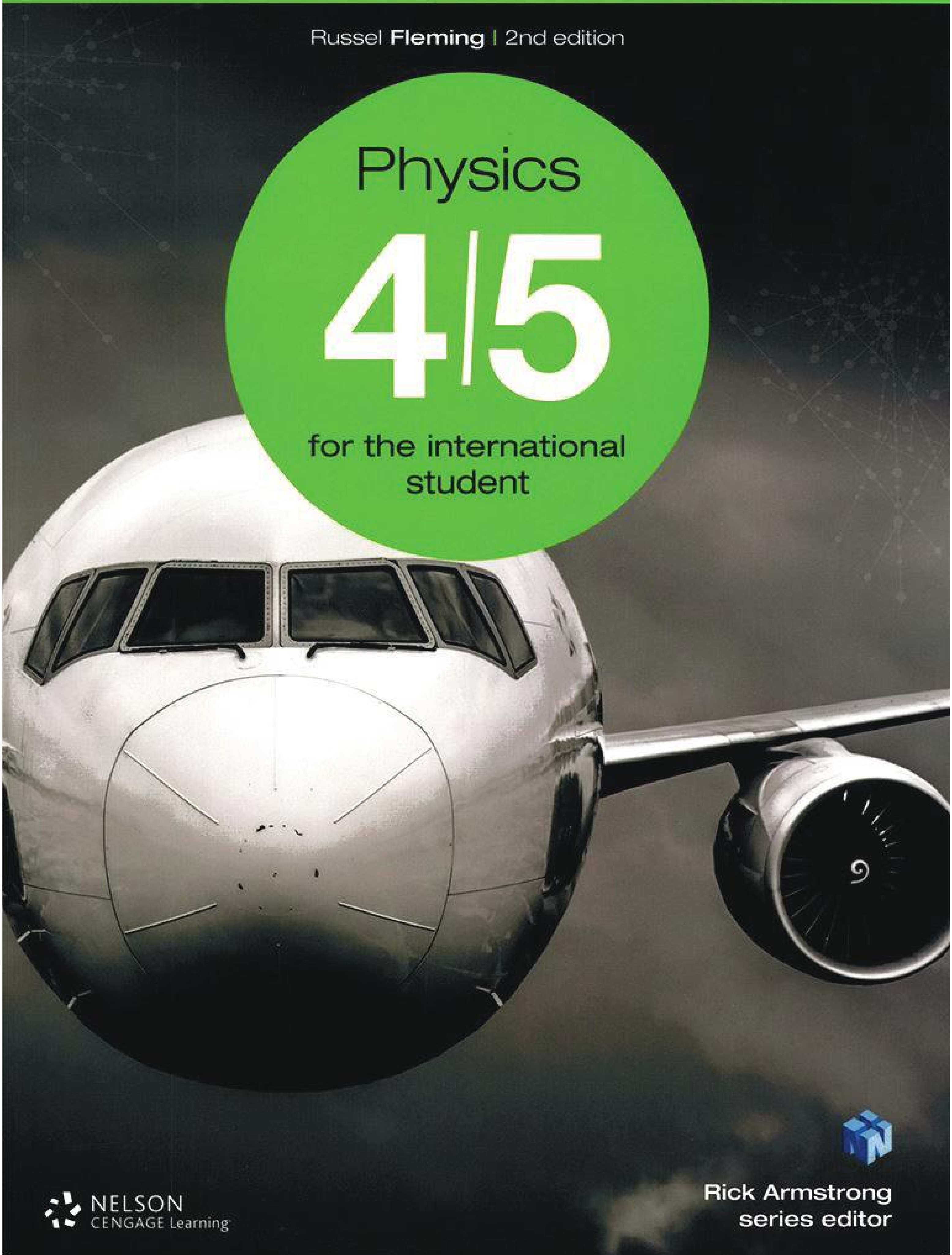
Russel **Fleming** | 2nd edition

Physics 4/5 for the international student



Rick Armstrong
series editor

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2nd Edition
Rick Armstrong
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How to use this series

The *Science for the international student* series provides students with a variety of engaging and stimulating formats for learning, understanding and immersion in both the Middle Years Programme (MYP) philosophy of the International Baccalaureate (IB) and the science content. The features of the student book have been specifically designed to support this and to deliver exciting content in a variety of ways.

Specific MYP features

Each unit begins with a unit opening page that specifies:

- the key concept that is covered in the unit
- the related concepts that are covered in the unit
- the Global Context of the unit
- the Statement of Inquiry
- inquiry questions, divided into factual, conceptual and debatable questions.

Key and related concepts

Each unit is based around one *key concept* of an enduring transdisciplinary nature and a small number of *related concepts* designed to help frame the unit in the minds of the students.

Global Context

Students will be encouraged to see science in diverse *global contexts*. These include historical developments in science; how scientific discoveries have affected medicine, space exploration, materials technology and nutrition; and the role of science in environmental sustainability and equitable access to resources in the developing world.

Statement of Inquiry

The *Statement of Inquiry* drives the unit and is strongly related to the units' concepts and context.

Inquiry questions

The inquiry questions are divided into factual, conceptual and debatable questions. Factual questions are related to the unit content, conceptual questions are related to the unit concepts and debatable questions are related to both and are designed to stimulate deeper thinking.

Performance assessment tasks

Opportunities for assessment tasks occur throughout each unit and these are each identified by a *performance assessment task* icon.

The *summative performance assessment task* associated with the Statement of Inquiry is identified at the beginning of each unit. The criteria assessed by the assessment task are also identified.

Approaches to Learning

Opportunities to develop and apply *Approaches to Learning* skills are identified by an 'ATL' icon. Teachers can use these prompts to discuss and reinforce learning strategies.

Investigation

Investigations challenge students to design and perform their own experiments either individually or in groups. Investigations are designed to satisfy criteria B and C.

Experiments

Experiments provide students with the opportunity to develop and practise their skills, by following processes and procedures, to discover information for themselves and to build a greater understanding of, and interest in, scientific concepts. Most experiments are designed to satisfy criterion, but some additional important experiments are also included (and the NelsonNet teacher website has further experiments).

Taking action

Taking action suggestions are identified by a 'TA' icon and are designed to satisfy the MYP requirements for service as action.

Other features

Review

Review boxes contain questions and break the content into smaller sections, allowing students to review what they have learnt so far.

Activity

Activity boxes reinforce or develop concepts and skills through short, fun and hands-on activities.

Weblinks

Weblinks are identified by an icon and direct students to exciting websites to further explore the world of science.

Unit questions

Unit questions conclude each unit. They include review questions sorted under the MYP assessment criterion A, levels 1–8. Reflection questions are included to review the concepts underpinning the unit, to encourage further consideration of the debatable inquiry questions, and at times to consider further lines of inquiry.

NelsonNetBook

The *Science for the international student* NelsonNetBook is an interactive ebook that can be used online or offline. It is compatible with interactive whiteboards, computers and tablets, with optional Web 2.0 functionality for class groups. Students can add highlights, annotations, audio and video clips, and weblinks, and teachers can use it to share their personalised version with the class.

Visit the NelsonNet portal at www.nelsonnet.com.au to find out more, register or log in if already registered.

NelsonNet teacher website

The NelsonNet teacher website contains further valuable advice, including draft MYP unit plans covering the first two pages of the revised MYP planner, and also a curriculum overview as required by the IB. Other resources include blackline masters (BLMs) containing possible further experimental work and classroom activities, ideas for further resources, and further advice relating to teaching in a conceptual way and for the use of the Approaches to Learning framework. Answers will also be provided for all questions, as well as a list of extra resources for each unit.

Contact your sales representative for information about access codes.

Introduction

To the student

We hope you will enjoy using this exciting student book, which has been designed to provide an up-to-date science experience around the principles of the new enriched Middle Years Programme (MYP) offered by the International Baccalaureate (IB). You are likely to already be an experienced MYP student, proud of being an *internationally minded* student, and familiar with the distinctive way MYP students work in science. These revised books provide a greater emphasis on the global contexts for learning in science, ranging from the challenge to provide better and more equal access to medicines worldwide, to considering global environmental challenges such as global warming. The books emphasise investigative and experimental work and expect you to work and think like a real scientist. As you will be well aware, the MYP is also about encouraging you to develop effective learning skills that will stay with you for life, and you will see in these books many suggestions to help you with this challenge. We wish you all the success possible with MYP Science and beyond.

To the teacher

We have reviewed our original series, published in 2010–2011, to take account of the innovative developments and improvements in the MYP. In this new edition, we have deepened our coverage of MYP principles within each unit. The units are now much more contextual and more explicitly driven by the Statement of Inquiry. As you will be aware, the IB has attempted to give schools more flexibility in their delivery of the MYP and there certainly is no 'correct' model of how to put the MYP into practice. For that reason, we feel we should explain some of our approaches to constructing our units.

1 **Conceptual framework:** We have closely followed the suggested framework but have added a small number of extra related concepts that will be useful to teachers and will allow coverage of the US cross-cutting concepts. We have also used concepts from other subjects when we felt their use would enhance the unit. Importantly, we accept that the key to teaching conceptually lies in appropriate classroom practices. To help this practice, we have included activities and questions to help strengthen students' understanding of the conceptual framework as well as some further guidance in the teacher materials.
2 **Content:** We have included academically challenging content that will provide an effective transition to higher study in the Diploma Programme (DP) or in other national systems. This content should also help teachers meet the requirements of local curricula or prepare for other examinations, such as IGCSEs. We have covered all the expected content for MYP Sciences e-examination in Books 4/5. Some of this content is also covered in more detail in Books 1, 2 and 3. We have ensured that the scope and sequence of our MYP Books 1–5 is well thought out and offers a coherent framework for the development of deep understanding based on the big unifying concepts in science.
3 **Global Contexts:** The development of the Areas of interaction into the Global Contexts is very liberating and opens the door for much more creative uses of contexts in the planning of MYP units. To take advantage of this potential, we have associated the Global Context chosen for the unit with a more specific 'exploration into' statement. This 'exploration into' feeds clearly into the Statement of Inquiry for each unit. This has helped us to make the science content up to date, interesting and relevant to the real world.

4 **Statements of Inquiry:** We have written simple and clear Statements of Inquiry that are understandable to students and to teachers. We have been flexible in relation to trying to build all the chosen concepts into the Statements of Inquiry. Our priority has been to ensure that the Statement of Inquiry is easy to understand, has a conceptual feel and, importantly, relates to the chosen Global Context.

5 **Assessment tasks:** Most science units will require more than one summative performance assessment task because it is artificial to try to bring together a number of the sciences criteria in one task. Therefore, most units include assessments relating to investigation work (criteria B and C), a performance-type task relating to the impacts of science (criteria D) and end-of-unit questions to assess criterion A. At the beginning of each unit, you will see a summative performance assessment task that relates closely to the Statement of Inquiry. We have given this task the most authentic performance nature possible. Other performance assessment tasks are included in each unit that can be used summatively or formatively. We expect that not all of the assessment suggestions will be used for summative purposes.

6 **Approaches to Learning:** We are very impressed by the revised Approaches to Learning framework based on the ten clusters of ATL skills. We understand that the effective implementation of ATL is a whole-school challenge but have made suggestions for when teachers can explicitly introduce these skills and dispositions, both as part of summative assessment tasks, and also more generally in their daily teaching.

7 **Service learning:** We have also suggested a possible service learning activity (labelled Taking action) for each unit.

The NelsonNet teacher website contains draft MYP unit plans, a curriculum overview, BLMs for experimental work and classroom activities, ideas for interdisciplinary tasks, further resources and advice for using the ATL framework, and answers to all questions.

We realise there may seem to be an inherent conflict between the idea of teachers working in a creative and collaborative way to produce MYP units of work and the use of a textbook. Schools will use this book in different ways. Some new schools might find it an invaluable stepping stone to getting a MYP Sciences programme up and running. Others may use it to enhance their existing courses. We encourage you not to use these books the way traditional textbooks have been used. Be creative, add to them, choose the bits you like, encourage the students to interact with them. They are there to help students in their deep learning of science, to encourage their interest and motivation. We hope the availability of materials of this kind will make your life as the teacher a little easier and give you more time to focus on the actual teaching and learning. Enjoy them.

Rick Armstrong (Series editor)

UNIT 1

LIGHT AND SIGHT IN FOCUS

KEY CONCEPT
Global interactions

RELATED CONCEPTS
Equity

Consequences

GLOBAL CONTEXT
Fairness and development – an exploration into inequality of access of different groups to prevention and treatment of blindness.

STATEMENT OF INQUIRY
Visual impairment and blindness have enormous implications for quality of life, and opportunities for their treatment vary globally.

INQUIRY QUESTIONS

FACTUAL
1 How does light travel?
2 How does the eye work?
3 How do glasses address vision problems?

CONCEPTUAL
4 How do mirrors and lenses work?
5 How do we perceive colour?
6 What are the different ways people can lose their sight?
7 What are the possible treatments for vision loss?

DEBATABLE
8 Is the developed world doing enough to support the treatment of diseases in developing countries?
9 Why are women more likely to be blind than men?

Introduction

ATL

REFLECTION: SHOWING SELF-AWARENESS OF YOUR LEARNING

Why might it be a good idea to consider what you expect to learn before you actually start the unit? What might you learn?

Our eyes allow us to see things. If we cannot see well, it takes a lot more effort to look after ourselves and others. Think about the consequences of being blind for your everyday life. What effect would blindness have on people of different age groups? How hard would it be to look after a baby if you were blind? Imagine that you have an eye disease and live in a remote area a long way from a doctor. What would you do? Would you stay where you are and suffer, or would you seek out help in a faraway location that you might not be able to afford?

In this chapter you will learn about how your sight works, the sight problems people can have, how the level of development of society can influence sight impairment, and what is being done globally to reduce sight impairment.

SUMMATIVE PERFORMANCE ASSESSMENT TASK

CRITERION D

Helping the vision impaired

At the end of this unit, you will be asked to contribute to the organisation of an event, such as World Sight Day, that seeks to improve conditions for people with sight problems. You will be asked to do further research on one of the sight issues caused by disease, diet or hygiene mentioned in this unit. Highlight how scientific knowledge is being used in the search for solutions, but also discuss the other factors involved in finding solutions to these sorts of problems. You will need to discuss the final organisation with your teacher, but one possibility is that the class produces a website.

FIGURE 1.1 A vision-impaired boy uses a white cane while walking with his father.

For a report on the costs and benefits of ending avoidable blindness, go to http://mypphys45.nelsonnet.com.au and click on **investing in vision**.

Blindness across the globe

Children who cannot see properly have difficulty learning to read and write. Adults with vision problems cannot work to their full potential. Imagine an office worker who can no longer read documents, a mechanic who cannot see up close in a car engine, a traditional artist who can no longer see colours properly, or a parent who cannot respond to her baby's non-verbal communications.

Many people are born with vision problems. Many develop difficulties because of environmental conditions (excessive sunlight, poorly lit work conditions) or the effects of bacterial infections and parasites.

Low-income communities are much more likely to suffer eye health problems, especially women. The World Health Organization estimates that about 269 million people have **low vision**. Eighty per cent of low vision and **blindness** is preventable or can be treated. By 2020, the number of blind people worldwide will have risen from 45 million to 76 million. There are 40 million blind people living in countries where people have low incomes. Women make up about 68% of all blind people, but this is much higher in low-income communities.

In this unit, you will learn about light, the behaviour of light and the structure of the eye, to help you understand what it means to have a **vision impairment**.

ACTIVITY Family life role-play

INTRODUCTION

People do not exist in isolation. They are always connected to other people in families, extended families and friends, and to the community through organisations, such as hobby and sports clubs, health centres, schools and their workplaces. What difference do you think it would make to a child, their family, neighbourhood, community and the wider world, if the child were blind? What extra resources would a family need to ensure the child grows up well?

For additional resources that give background and people's stories, go to http://mypphys45.nelsonnet.com.au and click on **family life**.

YOU WILL NEED

- A group of four people
- Video recorder

WHAT TO DO

1 Develop a script for a 4-minute role-play related to a scenario from family life.
2 Decide what kind of life the family leads. Is it similar to yours? Or, for example, does the family live in a small, rural village where they have ten meals a week?
3 Allocate roles to father, mother and two children. What ages are the children? Are they both the same gender or is one a boy and one a girl? One person is blind. You decide which person.
4 What scene from family life will you role-play? You could role-play preparing a meal, going to school, going to work, studying, or going for a trip on a bus or train.
5 Perform and record the role-play.

WHAT DID YOU DISCOVER?

How did the role-play help you to understand the way family members relate to each other when one member is blind?

ATL

COLLABORATION

Role-plays are a valuable form of learning and developing teamwork skills. Did you show good listening skills and empathy towards other students during this task?

Understanding light

A **luminous** light source produces light and sends it through space and time. **Rays** of light travel away from the source in straight lines. You can see this when you look at the light from a torch, a spotlight or the Sun coming through gaps in the clouds (Figure 1.2).

If light did change direction as it travelled through the air, you would be able to see around corners, and shadows might not exist.

FIGURE 1.2 Light rays from the Sun travel in straight lines.

Types of light source

There are two main types of light sources. An **incandescent** light source produces light as a result of heating. Heating causes electrons in the source to move randomly in all directions, sometimes crashing into each other. The random accelerations and decelerations of these electrons cause light radiation to be emitted.

A **luminescent** light source produces light as a result of absorbing other forms of energy. Generally, all the atoms become excited because of this energy absorption. They emit characteristic light radiation as they de-excite back to their original energy state (Figure 1.3).

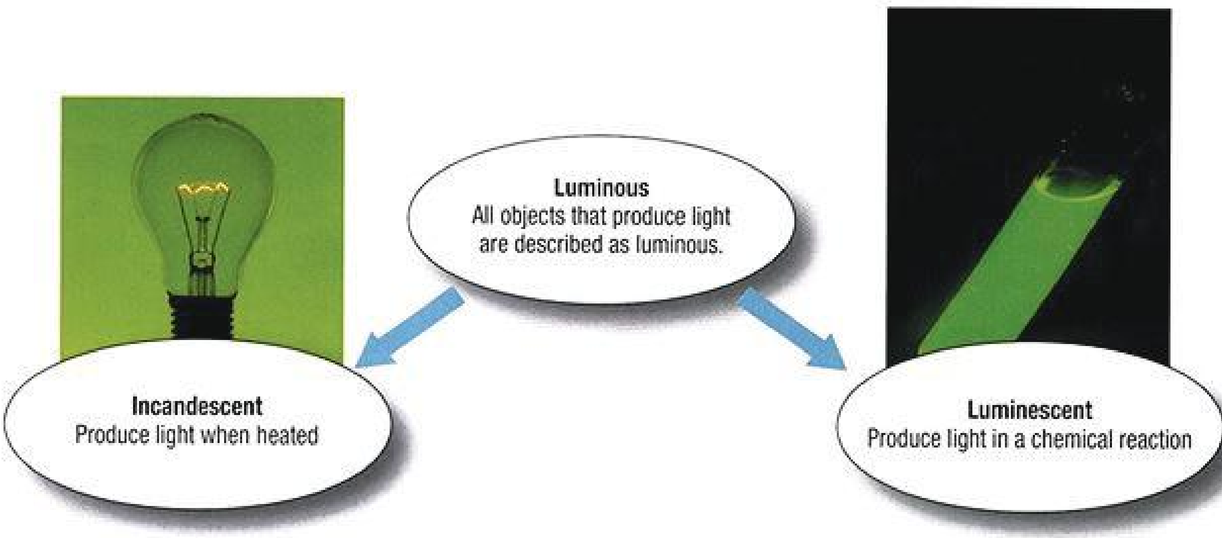

FIGURE 1.3 Luminous light sources are incandescent or luminescent.

FIGURE 1.4 A nearly new moon. The bright crescent is reflected sunshine. The less bright remainder is reflected earthshine.

Other sources of light

Most light that reaches our eyes comes from non-luminous objects that do not, themselves, produce light. The Moon is a good example of a non-luminous source. It reflects light from the luminous Sun towards us. We see moonlight, which is actually reflected sunshine.

Earth is also a non-luminous light source. During the phase of the Moon called the **new moon**, it is sometimes possible to see the outline of the Moon in the sky. Earth is reflecting the sunshine onto the Moon, and this sunshine is returned as **earthshine** (Figure 1.4).

Astronauts and cosmonauts in a space station can see Earth for the same reason they can see the Moon – the Sun's rays reflect off Earth towards them.

Blocking off light

A **transparent** material allows light through. **Opaque** materials do not let light through. Light bounces off these materials in the process called **reflection**. Light is also absorbed by these materials. The absorbed light is not reflected.

What is light?

A model for light

We do not know what light is, only how it behaves. In fact, to ask 'What is light?' is not as useful as to ask 'How does light behave?'. We can then use some familiar ideas, images or models to explain light phenomena.

Sometimes light behaves like a wave – this is the wave model of light. Sometimes light behaves like a particle – this is the particle model. And sometimes light seems to act as a particle that is simultaneously acting as a wave (wave–particle duality or **photon** model).

The electromagnetic radiation model of light

A wide variety of rays exhibit wavelike behaviours. These include gamma rays, X-rays, the colours we see, microwaves and radio waves. All of these rays have two parts – an electric part and a magnetic part – that work at right angles to each other as well as to the direction in which the ray travels. A wave that has these characteristics is called an **electromagnetic wave** (Figure 1.5). The **wavelength** of the wave is the distance between repeating parts of the wave.

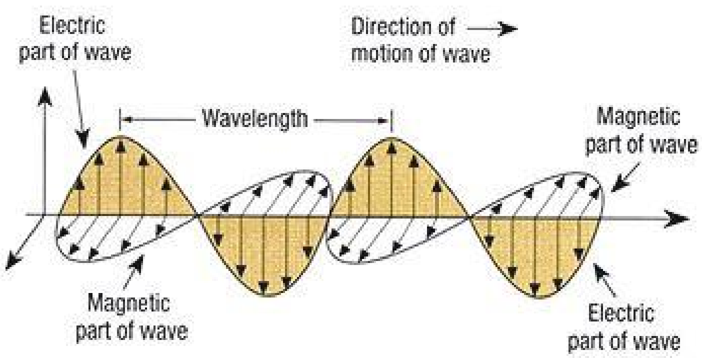

FIGURE 1.5 An electromagnetic wave travels at right angles to both the electric and magnetic parts of the wave. The wavelength is the distance between repeating parts of the pattern.

The **electromagnetic spectrum** (Figure 1.6) is the whole range of electromagnetic waves. Gamma rays have very small wavelengths but very high energies, and radio waves have long wavelengths and relatively low energies. Visible light is in between. The rate at which the wave repeats itself, the **frequency**, increases with energy.

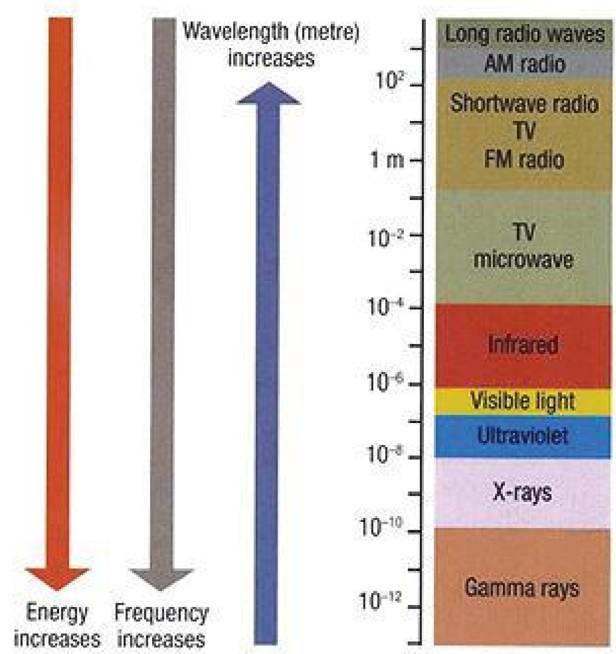

FIGURE 1.6 The electromagnetic spectrum

REVIEW

1 Copy and complete the following table.

Number of people in the world with low vision	
Percentage of people with low vision whose vision problems are readily treatable	
Number of people whose vision problems are preventable or treatable	
Number of blind people	
Percentage of people with low vision who are blind	
Number of blind people who live in countries where there is low income	
Percentage of all blind people who live in countries where there is low income	

2 Knowing that light travels in straight lines, sketch a diagram to show how your shadow is formed when you are standing in sunlight.

3 Outline why the formation of shadows is possible if light travels in straight lines.

4 Match each word with its meaning.

Luminous	Produces light by means other than heat
Transparent	Produces light when it gets hot
Opaque	Produces light
Incandescent	Does not let any light through
Luminescent	The bouncing of light off a surface
Reflection	Lets light through easily

5 Label these statements as true or false. Where a statement is false, rewrite it to make it correct.

- **a** Stars are luminescent.
- **b** Rocks are opaque.
- **c** Black objects let light pass right through them.
- **d** If there were no Sun or other stars, the Moon would be invisible.
- **e** Light behaves like a type of wave.
- **f** Gamma rays are the only part of the electromagnetic spectrum that we can see.

6 Is earthshine the same as or different from sunshine? Explain.

7 State the names of three different types of waves from the electromagnetic spectrum.

Bending light

Regular reflection

Light always bounces off at the same angle at which it strikes the surface. Some materials have very polished surfaces and act as mirrors. Rays that come in parallel are reflected as parallel rays. This predictable, **regular reflection** enables us to see an **image** or picture in the mirror.

Figure 1.7 shows a ray coming towards a reflecting surface. This is the **incident ray**. The ray that leaves the surface after being reflected is the **reflected ray**. The dotted line at right angles to the surface is the **normal**. Thus, the **angle of incidence**, $i°$, is the angle between the incident ray and the normal; similarly, the **angle of reflection**, $r°$, is the angle between the reflected ray and the normal.

The law of reflection states:

angle of incidence ($i°$) = angle of reflection ($r°$)

Scientists still send laser signals to a set of mirrors left behind on the Moon by Neil Armstrong and Buzz Aldrin in 1969 (Figure 1.8) and record the time delay for the return journey. This gives them precise measurements of the distance from Earth to the Moon, the rate of the Moon's rotation, and gravity throughout the universe.

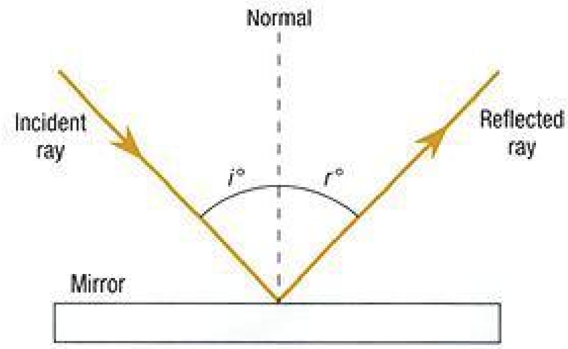

FIGURE 1.7 Regular reflection. Incident parallel rays are reflected parallel because $i° = r°$.

FIGURE 1.8 Mirrors on the Moon. Laser signals are sent from Earth to the Moon and reflected.

Go to http://mypphys45.nelsonnet.com.au and click on **Moon landing**. Read about the cutting-edge science experiment left behind in the Sea of Tranquility by Apollo 11 astronauts that is still running today.

Diffuse reflection

Most opaque materials have irregular surfaces so that the light reflects in many directions. This **diffuse reflection** is the reason why we can see surfaces from many different positions. Incoming parallel rays bounce off the bumpy surface of the material and are scattered (Figure 1.9).

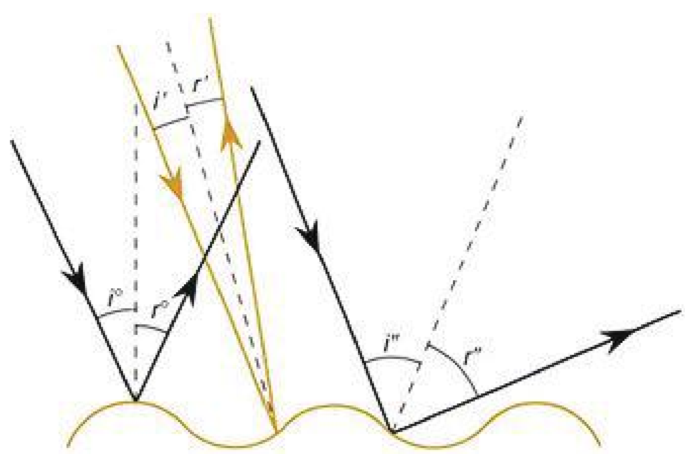

FIGURE 1.9 Diffuse reflection. The rays scatter because $i° = r°$ on a bumpy surface but the normal changes at each different point.

EXPERIMENT 1.1

Travelling light: reflection

AIM

To use a light box to study the reflection of light

MATERIALS

- Light box kit
- Sheets of plain white paper
- Ruler and sharp pencil

PROCEDURE

PART A: MAKING A BEAM INTO A RAY

1 Connect the light box to the power supply. Dim the lights so that you are working in a partially darkened room.

2 Place a sheet of white paper on the table in front of the light box and adjust the box so that you have a beam of light with parallel sides.

3 Use one of the plastic slides to make a single ray of light. This ray of light should be clearly visible on the white paper (Figure 1.10).

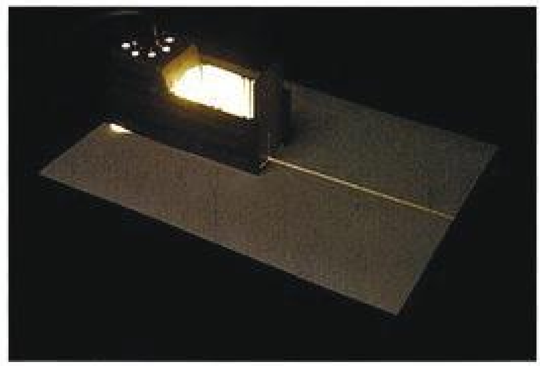

FIGURE 1.10 A very narrow beam of light

PART B: REFLECTION FROM A FLAT (PLANE) MIRROR SURFACE

4 Predict what will happen to the light when you send a ray to the plane mirror, but not 'head-on'. Try this out. Was your prediction correct?

5 Trace the shape of the mirror onto the paper and use dots to mark the positions of the incident ray and the reflected ray.

6 Remove the mirror. Rule in the incident ray and the reflected ray. Insert arrowheads to show the direction of the rays.

7 Now draw in a normal line (at right angles to the mirror), as shown in Figure 1.11.

8 Measure the angles of incidence and reflection.

9 Repeat for a range of angles of incidence.

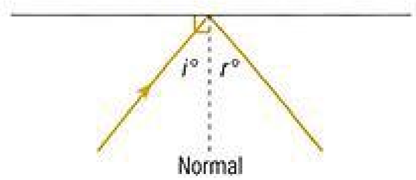

FIGURE 1.11 Angles of incidence and reflection

CONCLUSION

How close are your results to confirming the law of reflection?

angle of incidence ($i°$) = angle of reflection ($r°$)

EVALUATION

Discuss the errors in this experiment. How could they be reduced? Why do you think we measure the angles relative to a normal line, not relative to the side of the mirror?

Refraction

When light rays reach a boundary with a new transparent material, they travel into the new material but change direction. This is called **refraction** (Figure 1.12). The angle between the normal and the refracted ray is called the **angle of refraction, *R*°**.

When light travels through a glass block, rays are refracted at the boundary of the air and the block and then again at the boundary of the block and the air.

You can see refraction effects when you look at someone who is standing in water (Figure 1.13).

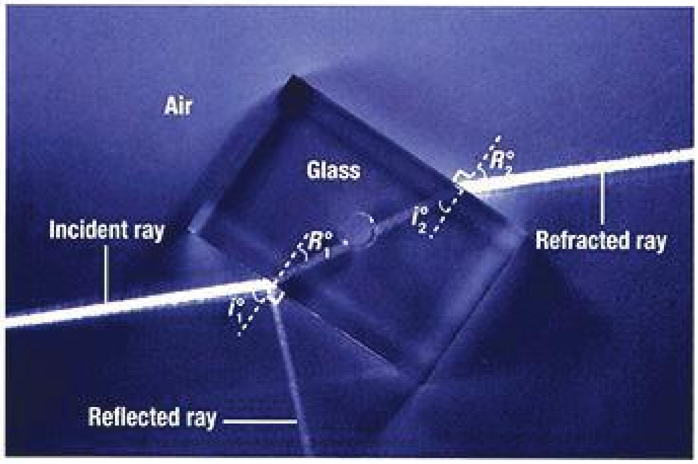

FIGURE 1.12 Rays are refracted at the air to block boundary and at the block to air boundary. i° = angle of incidence; R° = angle of refraction.

FIGURE 1.13 Refraction between air and water changes the apparent shape of the person's legs.

EXPERIMENT 1.2 Refraction of light

PERFORMANCE ASSESSMENT TASK CRITERION C

AIM

To use a light box to study the refraction of light

HYPOTHESIS

Write and explain a hypothesis to predict what will happen to light rays when they strike a rectangular block of transparent material.

MATERIALS

- Light box kit
- Sheets of plain white paper
- Ruler and sharp pencil
- Transparent rectangular block
- Triangular **prism** (for Extension)

PROCEDURE

1 Place a fresh sheet of white paper on the table. Place the rectangular block onto the paper in the path of the light ray. What happens to the light? Take care to observe what happens at the front and the back of the block (see Figure 1.12). You should see both reflection and refraction.
2 Trace around the rectangular block so that its shape is recorded on your paper. Then mark in dots to show the positions of all light rays.
3 Remove the block and complete and fully label the diagram. You will need to join up the points at which the light entered and left the block. Use arrowheads to show the direction of the rays. You should have a diagram that looks like Figure 1.12.
4 Draw in the normal lines at the points where the light both enters and leaves the block.
5 Repeat the procedure with a variety of angles of incidence of light entering the block.

>

RESULTS

1 Make a table to show how the angle of refraction varies as the angle of incidence is changed (as the ray enters the block).
2 Plot a graph of angle of refraction versus angle of incidence.

CONCLUSION

1 Describe in general what happens to the ray of light at the boundary where it enters, and at the boundary where it leaves the block.
2 Include a comment on any reflection you see. Does the amount of reflection remain the same with different angles of incidence?
3 Describe qualitatively any general trends in the relationship between the angle of incidence and the angle of refraction, as shown by your graph.
4 Evaluate the validity of your hypothesis.

EVALUATION

Discuss the validity of this experiment and describe possible improvements and extensions to the investigation.

EXTENSION

1 On a new sheet of white paper, place a triangular prism head-on in the path of the light. Trace the shape of the prism and mark in what happens to the light rays. Label your diagrams.
2 Place another sheet of white paper on the table and try rotating the triangular prism. Record what happens in each position.

CONCLUSION

1 State the main difference(s) between the way a ray of light behaves when it meets a transparent triangular prism and a transparent rectangular prism.
2 What happens to the ray of light that emerges from the triangular prism as the angle of the incident ray from the light source changes?
3 Did more light pass through or reflect from the transparent rectangular block than from the triangular prism? On what evidence did you decide this?
4 Did the light ray ever stop travelling in straight lines? Comment.
5 Why do you think that colours appeared in the rays when they emerged from the triangular prism?

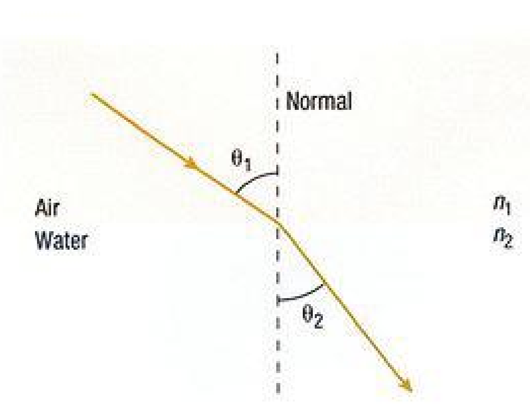

FIGURE 1.14 Light refracting at the air to water boundary

The Descartes–Snell law of refraction

The law of refraction is a relationship that connects the angle of incidence, θ_1, with the angle of refraction, θ_2. In line with other definitions, the angle of refraction is the angle between the refracted ray and the normal. The Descartes–Snell law is:

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{n_2}{n_1}$$

where n_1 is the **index of refraction** of the incident medium, in this case air, and n_2 is the index of refraction of the refracting medium, in this case water.

The index of refraction is a number that represents how much the material refracts light; that is, its unique **refrangibility**. The larger the index, the greater the refrangibility. The index of refraction of air is 1.00.

Law of refraction calculations

WORKED EXAMPLE

EXAMPLE 1

Calculate the refractive index for water in Figure 1.14 given the following information.

Angle of incidence = 55°

Angle of refraction = 38°

Write out the working before typing into your calculator:

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{n_2}{n_1}$$

$$n_2 = \frac{n_1 \sin\theta_1}{\sin\theta_2} = \frac{1.00(\sin 55^\circ)}{\sin 38^\circ} = 1.33$$

The refractive index of water is 1.33.

We can also do the reverse. We can predict an angle of refraction if we know the angle of incidence and the index of refraction.

EXAMPLE 2

Calculate the angle of refraction for a ray in air that enters a diamond at 45°. The index of refraction of diamond is 2.42.

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{n_2}{n_1}$$

$$\sin\theta_2 = \frac{n_1 \sin\theta_1}{n_2} = \frac{1.00(\sin 45^\circ)}{2.42}$$

$$\sin\theta_2 = 0.29$$

Angle $\theta_2 = 17°$ (use the inverse sin function on your calculator)

ACTIVITY

Re-plot your graph from Experiment 1.2 in the form of sin i° versus sin R°. If you have not used the sin function before, your teacher will give you an explanation and show you how to do the calculation on your calculator or your computer.

- Does this line agree with the Descartes–Snell law of refraction?
- Use your graph to calculate the value for the refractive index of the material you used.
- Again, evaluate your results.

INVESTIGATION 1.1 Refraction of light

PERFORMANCE ASSESSMENT TASK
CRITERIA B (I, II AND III) AND C

YOUR CHALLENGE

Use the simulation in the weblink to test one aspect of the Descartes–Snell law.

THIS MIGHT HELP

It can be difficult to design experiments to investigate different aspects of the Descartes–Snell law because of technical difficulties, such as measuring small angles and being able to accurately change the index of refraction. In this simulation you can easily change and measure several aspects of light, even its speed. Click on 'More Tools' to access more features of the simulation.

Carry out and write up your investigation following the guide in the Appendix on page 271 or as advised by your teacher.

Go to http://mypphys45.nelsonnet.com.au and click on **bending light simulation** to explore the bending of light between two media.

Lenses

A **lens** is a carefully curved transparent object that is able to refract light in a predictable way to form images. The lens in the eye is the eye's main image-forming device. There are two types of lenses. A **convex lens** is fatter in the middle than it is at the ends. A **concave lens** is thinner in the middle than at the ends (Figure 1.15). The lens in the eye is a convex lens.

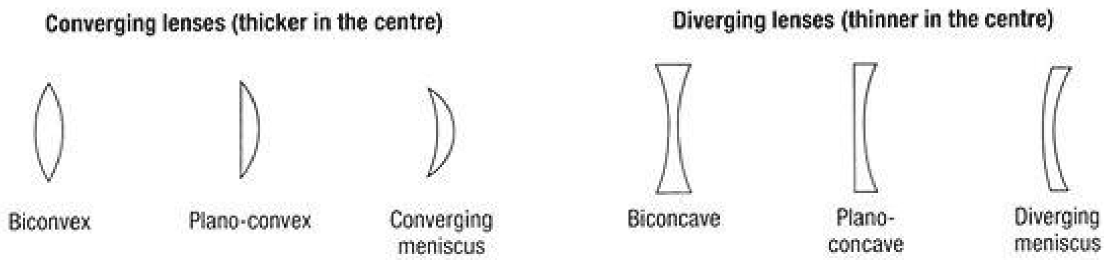

FIGURE 1.15 Convex and concave lenses

Physicists and chemists combine to develop sophisticated lenses to improve sight. Contact lenses are made of soft and hard plastic. Graded lenses, lenses that change colour and carefully shaped plastic lenses are used to replace damaged natural lenses.

Convex lenses

Artificial lenses are usually made from glass or plastic. Convex lenses are used to bring parallel rays to a point (**converge**), called the **focus** (F). The refracted rays really do pass through this **focal point** so the focus is called a **real focus** (Figure 1.16).

Go to http://mypphys45.nelsonnet.com.au and click on **lenses animation** to explore how lenses form images.

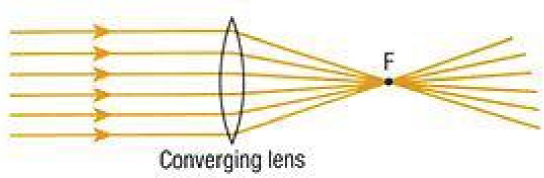

FIGURE 1.16 A convex lens converges parallel rays to a real focus.

Concave lenses

Concave lenses are used to **diverge** parallel rays away from the focus. The refracted rays do not really pass through this focal point – they only appear to – so the focus is called a **virtual focus** (Figure 1.17).

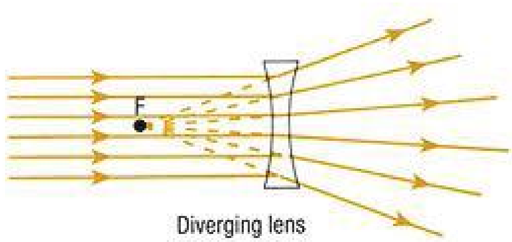

FIGURE 1.17 A concave lens diverges parallel rays from a virtual focus.

EXPERIMENT 1.3 Convex lenses

PERFORMANCE ASSESSMENT TASK
CRITERION C (I, II, IV AND V)

AIM

To use a light box to study the refraction of light through a convex lens

HYPOTHESIS

Write and explain a hypothesis to predict what will happen when you put a convex lens in the path of light rays. Place a fresh sheet of white paper on the table. Place the convex lens onto the paper in the path of a set of parallel light rays. What happens to the light?

MATERIALS

- Light box kit with convex lens
- Sheets of plain white paper
- Ruler and sharp pencil

PROCEDURE

1 Trace around the lens so that its shape is recorded on your paper.
2 Mark in a line that strikes the widest part of the lens at right angles. This is the **principal axis**.
3 Mark in a line through the centre of the lens that is at right angles to the principal axis. Label this axis LL' (Figure 1.18).
4 Direct a set of parallel rays so that the middle ray travels along the principal axis towards the lens. Mark in dots to show the positions of all light rays on their way towards the lens and after they have gone through the lens.

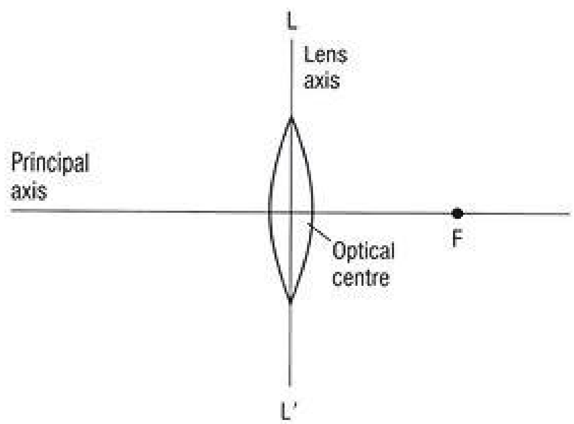

FIGURE 1.18 The axis system for a convex lens

5 Remove the lens and complete and fully label the diagram. Use arrowheads to show the direction of the rays. The point at which the rays converge on the side opposite the object is called the (real) focus or focal point. Label the focus F.
6 Use a single ray for this part. Place a sheet of white paper on the table. Mark in the principal axis and the lens axis. Mark an X about 2 cm above the principal axis and about 8 cm from the lens. Aim several different rays through the centre of the X towards the lens (Figure 1.19).

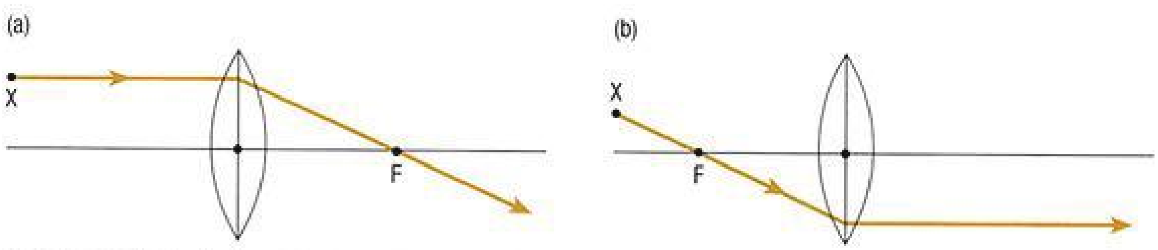

FIGURE 1.19 Following the paths of rays from point X, through a lens. (a) A ray parallel to the principal axis is refracted (by the lens) to pass through F. (b) A ray arriving through F is refracted parallel to the principal axis.

7 Trace the rays to the other side of the lens. Take particular notice of what happens to the ray that is sent parallel to the principal axis after it passes through X.

>

RESULTS

1 What happens to the rays coming from X just after they leave the lens?
2 Can you identify a point past the lens at which all the rays from X arrive? If you can, this is the real image of X. Call the point X′. (If the rays appear to diverge from a point on the same side of the lens as X, the image X′ is a virtual image).
3 How far from the lens is the image X′?
4 How far from the principal axis is X′? Compare this distance with the distance of X from the principal axis.

CONCLUSION

1 Explain why a convex lens is also referred to as a converging lens.
2 How does your data for this arrangement demonstrate a real image?
3 Discuss the validity of your hypothesis.

ATL

CRITICAL THINKING

Evaluating the validity of scientific experiments and discussing possible improvements to experimental methods is at the heart of effective scientific investigation.

EVALUATION

Discuss the validity of this experiment and describe possible improvements and extensions to the investigation.

EXTENSION

What happens to the position of the image (distance from lens and distance from principal axis) when you move point X parallel to the principal axis and closer to the lens?
What difference does it make if you use a fatter lens?

EXPERIMENT 1.4

Concave lenses

PERFORMANCE ASSESSMENT TASK
CRITERION C (I, II, IV AND V)

AIM

To determine how the image from a concave lens differs from that of a convex lens

PROCEDURE

Follow the procedure outlined in Experiment 1.3, except use a concave lens instead of a convex lens.

CONCLUSION

Compare the way convex and concave lenses work. How are their images different?

EXTENSION

Look through some concave and convex lenses. Relate what you see to the results of your experiment. Which of the images can you focus on a piece of paper?

REVIEW

1 Figure 1.20 shows two rays leaving a point X and travelling to a mirror surface MN. The rays strike the mirror surface at A and B. The reflected rays to the eye are not shown.
   a Copy the diagram and carefully draw in the reflected rays until they reach the eye.
   b Show that these two rays appear to come from a point, I, on the other side of the mirror from the object.

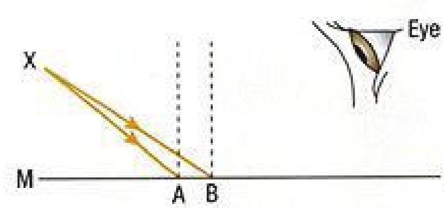

FIGURE 1.20 Rays from X strike the mirror surface MN.

>

2 Sketch and label a diagram to show how diffuse reflection works.

3 Figure 1.21 shows a light ray directed towards a transparent rectangular block. The ray exits the block from the opposite side. Show what happens to the light as it travels from air into the glass and then back into the air.

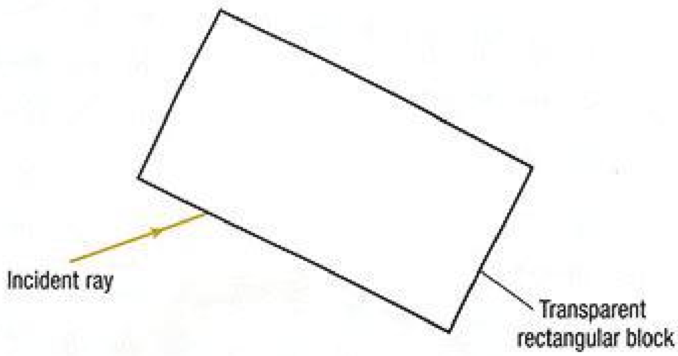

FIGURE 1.21 A light ray is directed at a transparent rectangular block.

4 Draw a diagram to show what happens to white light when it travels through a triangular prism.

  a State the order of colours that you see emerging from the prism.

  b Are the colours that you listed in part **a** really the only colours? Discuss.

5 Figure 1.22 shows four rays leaving an object. They are directed towards a convex lens. The position of the image of the object is shown on the opposite side of the lens. Show what happens to each ray as it travels to, and through, the lens.

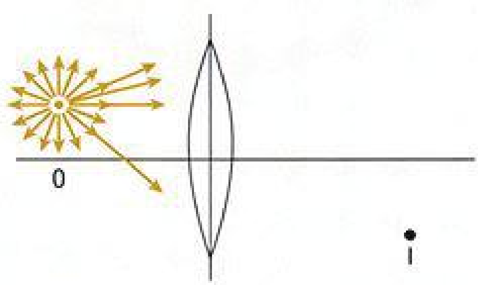

FIGURE 1.22 Rays leaving object 0 are directed towards a convex lens. The image is at I.

6 Construct an appropriate data table to place the following materials in order of refrangibility: diamond (refractive index 2.42), water (1.33), silicon (4.01), glass (1.49), air (1.0013)

7 A ray of light passing from a submerged waterproof flashlight is incident on the surface of the water (refractive index 1.33) at an angle of 30°. Calculate the angle of refraction.

8 A ray of light is sent through air towards a variety of transparent materials and then refracts inside the substance. Copy and complete the following table.

Angle of incidence (°)	Angle of refraction (°)	Index of refraction of material
35	26	
25		1.48
	18	2.42

The eye

Structure of the human eye

The human eye is shaped like a ball that is roughly 2.5 cm across (Figure 1.23). The eyes are set into the skull, and so are somewhat protected from damage. Eyes are one of nature's best optical instruments. They can focus on distant as well as near objects, in full sunlight and in very dim conditions.

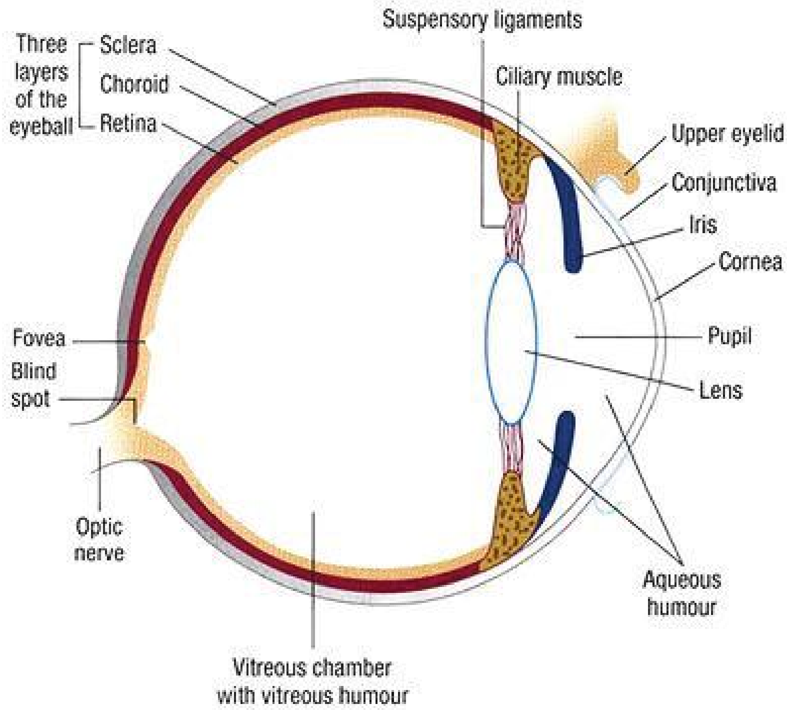

FIGURE 1.23 Structural features of the human eye

Receiving the light

Light enters our eyes from light sources, directly or indirectly, via reflection. Figure 1.24 shows light entering the eye through the curved, transparent **cornea**.

In the centre front of the eye, the **pupil** looks black because there is no light behind it. The coloured, muscular **iris** contracts in dim light to open the pupil and let more light in. The reverse happens in bright light. Did you know that the size of your pupils changes with your emotions? If you are happy or pleased, your pupils **dilate**. People can appear more attractive when their pupils are dilated.

Light passes through the cornea where about 80% of the refraction occurs, and then passes through the lens. This convex lens is a transparent, elastic tissue that can change shape slightly to form a clear image on the back of the eye; this is called **accommodation**. As light rays pass through the cornea and the lens, they refract and converge. The light then reaches the back of the eye, via the **vitreous humour**, and forms an upside-down image on the **retina** (Figure 1.24).

In accommodation, the **ciliary muscles** pull the lens tight or push it into a more spherical shape to improve the image. As we get older, the ciliary muscles lose their elasticity and the lens gets harder, so accommodation is more difficult.

Most vision problems are refraction related. They include short-sightedness, long-sightedness and astigmatism.

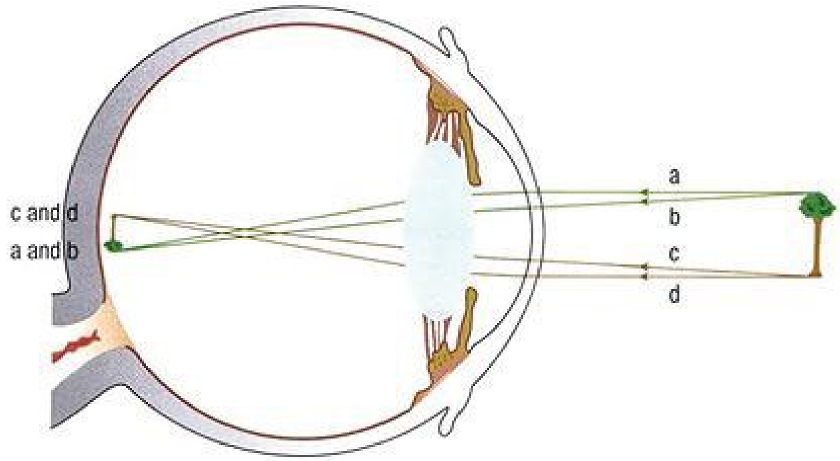

FIGURE 1.24 Focusing the light. Light rays refract at the cornea and the lens to form an image on the retina. The optic nerve carries the information to the brain.

Sending the image to the brain

The retina is a surface that is covered with millions of **visual cells** that are sensitive to light. The two sorts of visual cells, **rods** and **cones**, convert light into electrical signals, which travel to the brain along the **optic nerve** (see Figure 1.23). The retina is a thin layer of nervous tissue that acts as a screen. It does a lot of elaborate and sophisticated data-processing before the signals are sent to the visual-processing centre of the brain.

Blind spot

In each eye, there are no visual cells at the point where the optic nerve joins the retina. This is the **blind spot**. Light from a position within the view in front of you that hits this area cannot be seen. Fortunately, your brain puts the full picture together using information from your other eye.

Go to http://mypphys45.nelsonnet.com.au and click on **blind spot**. Follow the activity to find your blind spot.

Rods

There are approximately 120 million rods spread over the retina. Rods are highly sensitive to the intensity of light. They enable us to see things even when there is very little light coming from around us, such as at night. You will have noticed that when you move from bright light into the dark, it takes some time for your eyes to adjust – rods need time to recover after you enter a dark room.

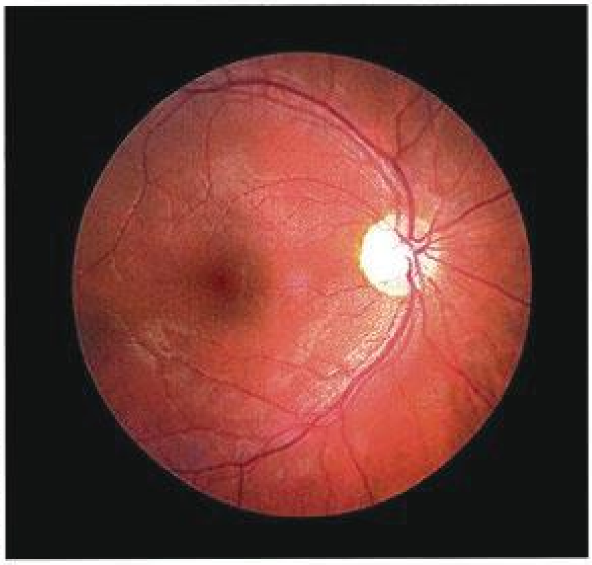

FIGURE 1.25 Inside the human eye: retina, optic nerve (bright) and macula (darker) with the fovea in the centre

Cones

Cones are responsible for colour vision and for precise vision. They are concentrated on the **fovea**, a small part of the retina about 0.2 mm in diameter, at the back of the eye above the optic nerve, in a direct line from the centre of the lens. The fovea lies at the centre of the **macular pit** (yellow spot), where light is most easily able to fall directly on the cones (Figure 1.25 and Figure 1.26).

There are approximately 6 million cones in an eye. They do not respond well to low light levels. This is why we do not see really precise images or colours very well at night.

Macular degeneration is a disease of the eye that robs it of the ability to see colour and the centre of the field of view.

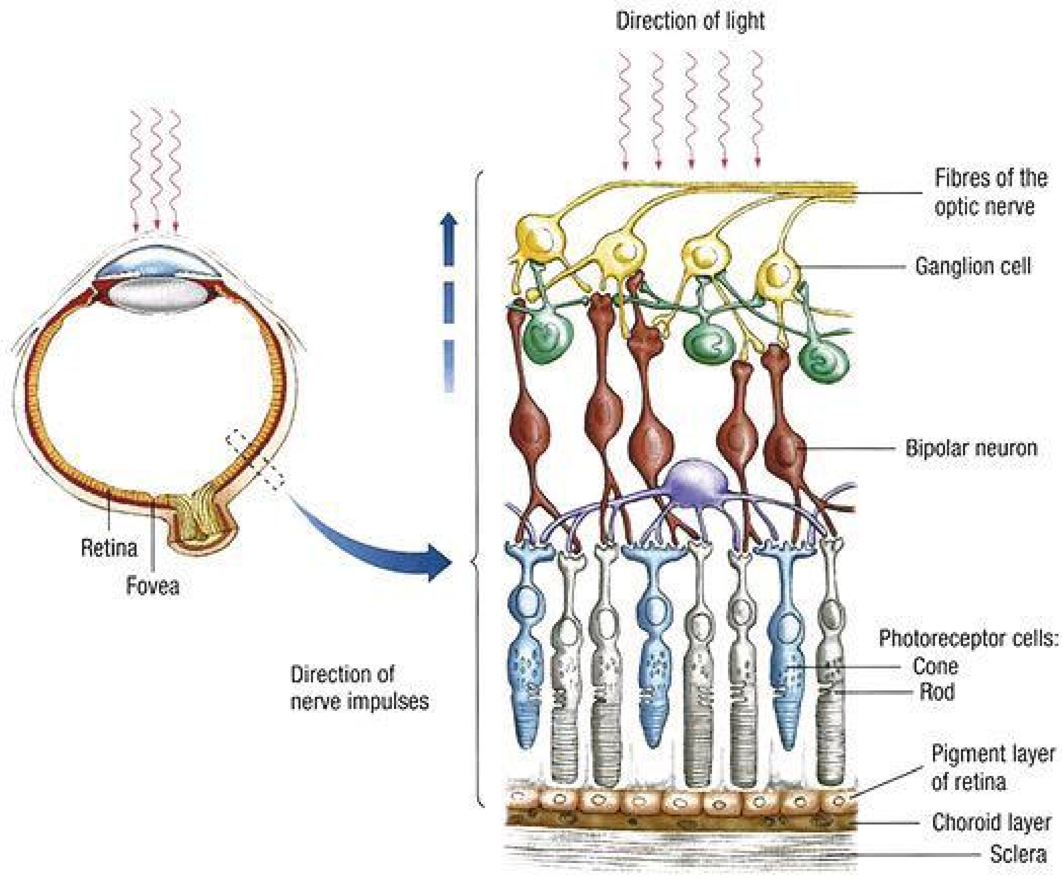

FIGURE 1.26 The retina. Light travels through layers of blood vessels, nerves and cells to rods and cones. In the macular pit and, especially, the fovea, it travels almost directly to cones.

Types of cones

There are three types of cone cells on the retina and they are sensitive to different colours of light – red, green and blue. Red light stimulates the red cones and yellow light stimulates the red and green cones (Figure 1.27).

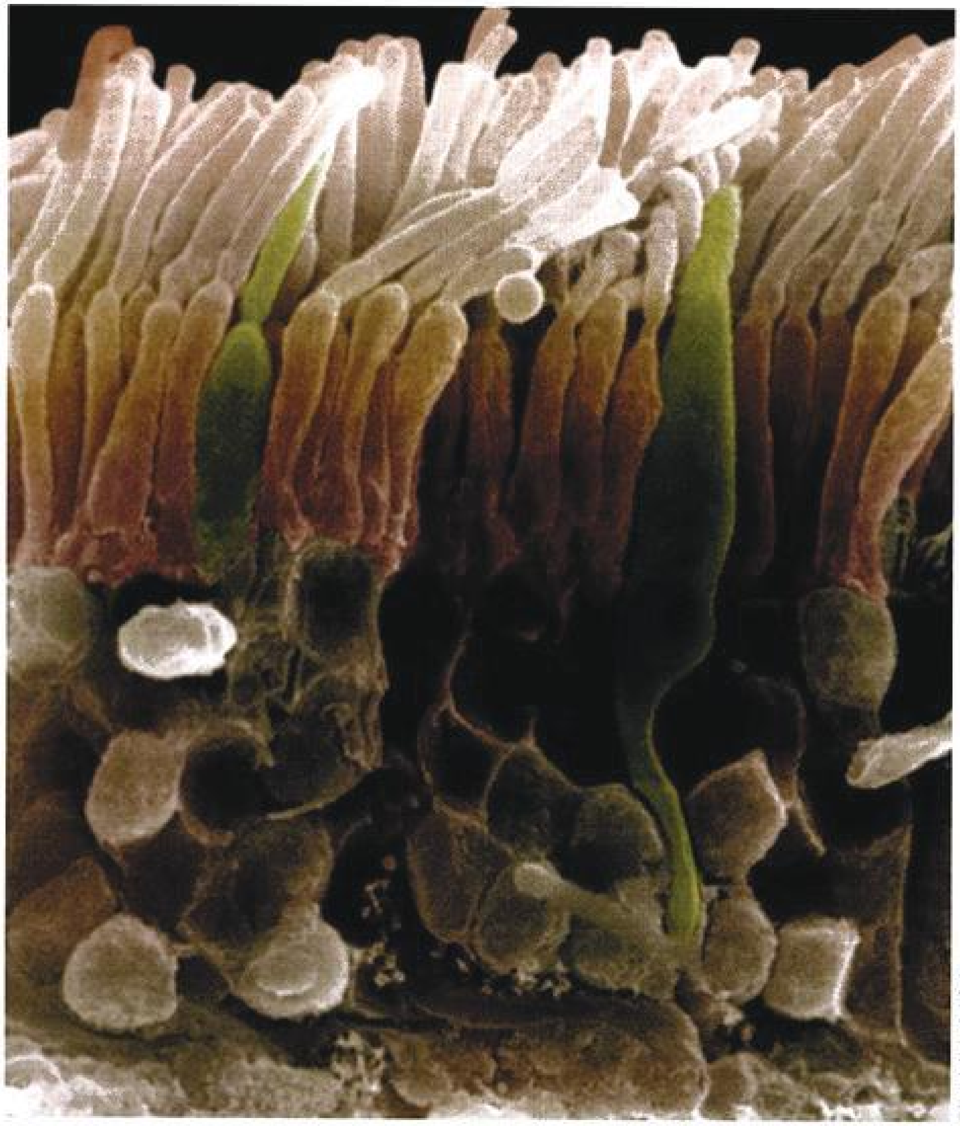

FIGURE 1.27 Highly magnified image of rods and cones in the human eye

ACTIVITY

Eye dissection

PROCEDURE

Carry out this dissection either via an online virtual eye dissection website or using the instructions given to you by your teacher to carry out a real dissection.

RESULTS

Identify all the features of the eye: cornea, pupil, iris, vitreous humour, lens, retina, fovea, optic nerve and muscles.

Go to http://mypphys45.nelsonnet.com.au and click on **eye dissection**. Follow the instructions to perform a virtual eye dissection.

ATL

TRANSFER: TOOLS TO IMPROVE MEMORY

A mnemonic, such as ROY G BIV, is a device to help memory. Alliteration or rhymes can also help such as 'Blue light bends the best'. Discuss why this helps memory. What other techniques can help memory? How are memory and learning related?

Colour

We see coloured things all around us. We talk about primary colours both in paints (pigments) and in the mixing of lights. Yet the primary colours in painting are not the same as those when mixing stage lighting. To understand this difference, we need to consider **additive primary colours** and **subtractive primary colours**.

White light (that is, the visible part of the electromagnetic spectrum) is made up of many colours. A prism can break up white light into all the colours humans are able to see. This is the **colour spectrum**, which can most naturally be seen in a rainbow. These are the colours you saw in the experiment with the triangular block (Experiment 1.2 Extension).

Different colours refract by different amounts. They separate as they leave the prism. Red light bends the least, violet light the most.

These spectral colours are defined by their appearance on a white screen. They are red–orange–yellow–green–blue–indigo–violet (ROY G BIV). Spectral colours cannot be separated into more colours. They appear to be fundamental colours for the human brain (Figure 1.28).

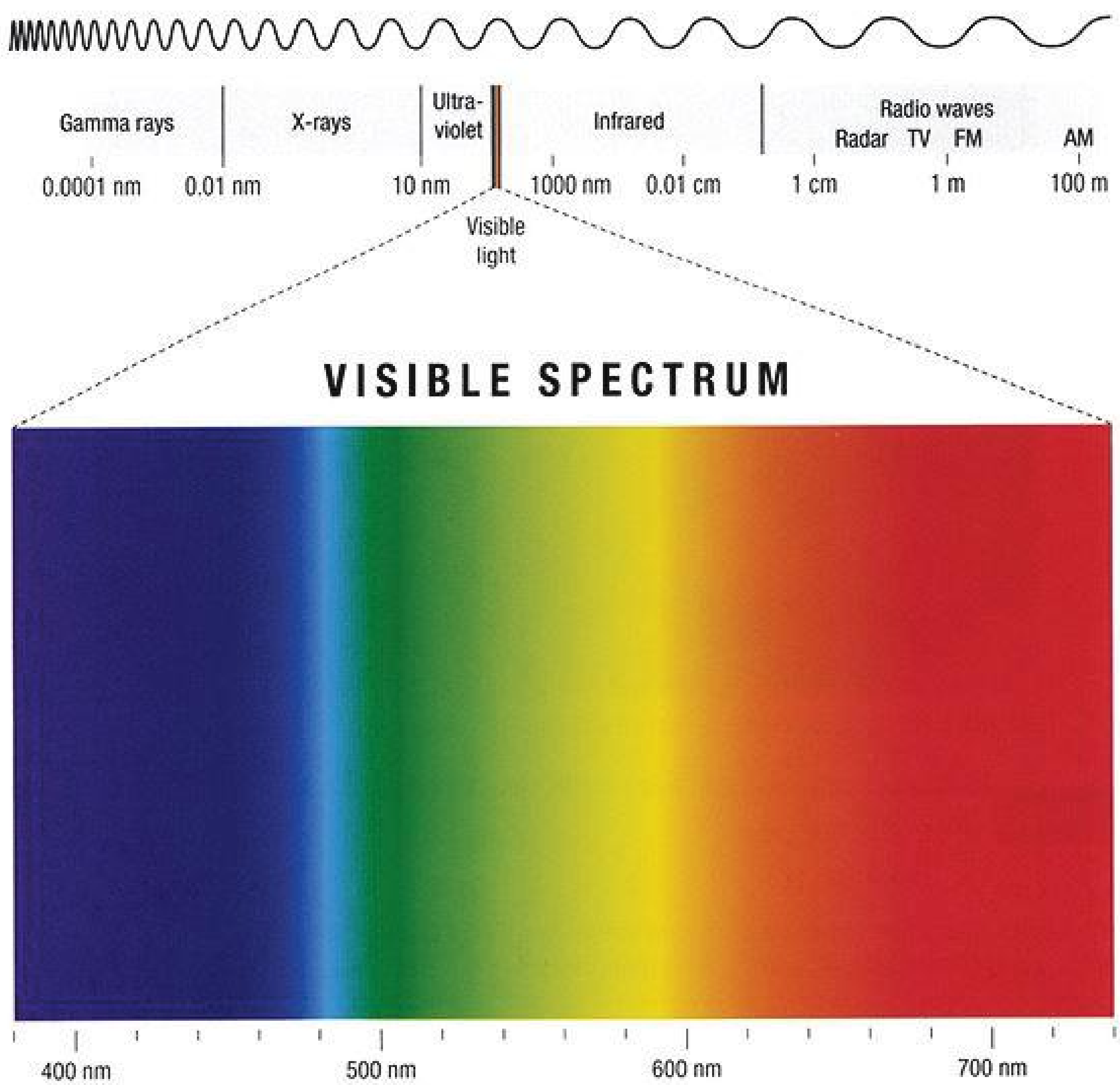

FIGURE 1.28 The spectrum of white light. Different colours refract by different amounts. They cannot be further separated into other colours.

Full circle rainbows

A rainbow is not a bow but a full circle. From the ground, the bottom part of the circle is cut off at the horizon. From up in the air, you can see the full circle (Figure 1.29).

Rainbows can only be seen when sunlight from behind us hits drops of rain water in front of us. The light travels in the water droplets, refracting, reflecting and refracting again as it is returned towards us. This spreads the white light into its component colours. The rainbow you see is built up from the tiny rainbows created by billions of raindrops.

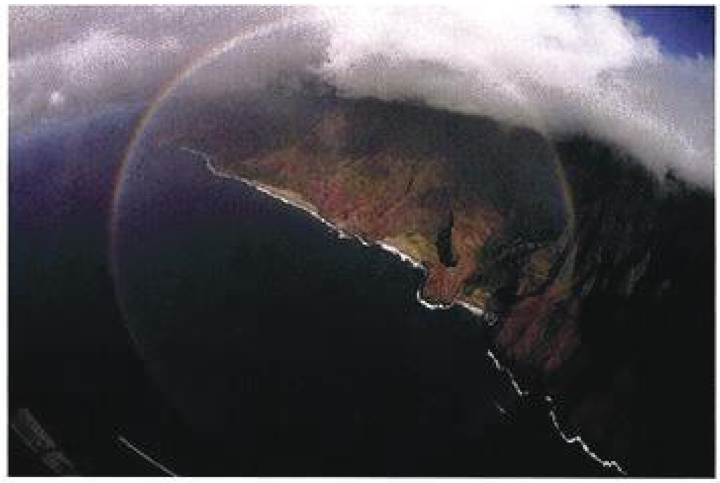

FIGURE 1.29 A rainbow circle can be seen through an aeroplane window.

Colour mixing

Additive mixing

Understanding colour is not at as simple as understanding the spectrum of visible light. You can produce red, blue and green light by passing white light through red, blue and green filters respectively. Red, blue and green are called the additive primary colours. When these three colours are shone on a white surface they add to produce white light (Figure 1.30).

If two of these colours overlap, they produce magenta, cyan or yellow light (Figure 1.31). Magenta, cyan and yellow are **secondary colours**.

When a primary colour is mixed with a secondary colour to produce white light by additive mixing, we say that the colours are **complementary colours**. Primary blue and secondary yellow are complementary colours because they produce white light (Figure 1.32).

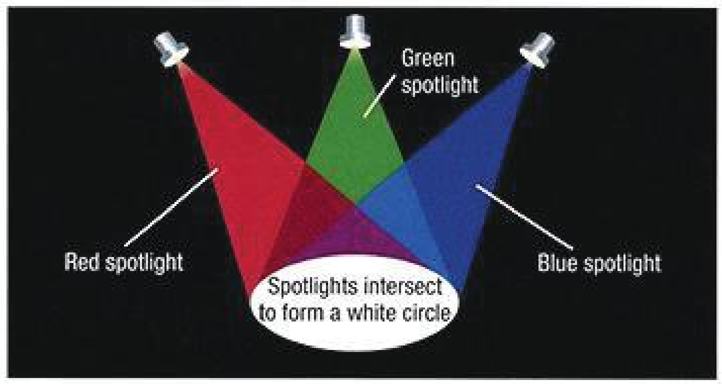

FIGURE 1.30 The primary colours, red, blue and green, add to make white light.

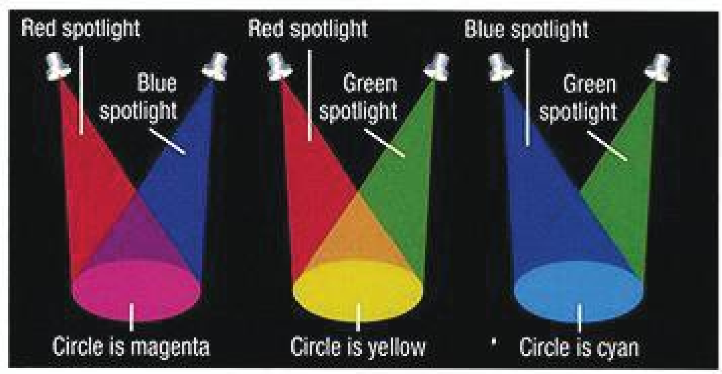

FIGURE 1.31 The secondary colours of light can be created.

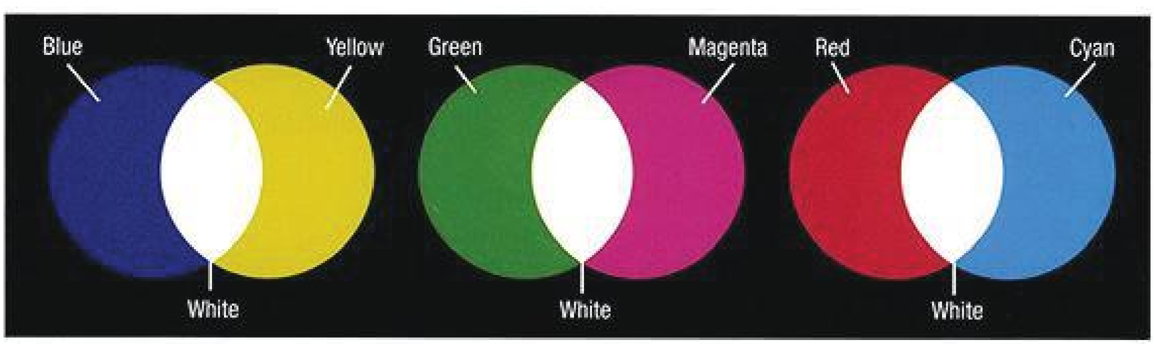

FIGURE 1.32 Complementary colours, such as primary blue and secondary yellow, add to produce white light.

Subtractive mixing

Subtractive mixing occurs when secondary colours overlap either through the mixing of pigments or when two coloured filters are put together. In these cases, magenta, cyan and yellow are called the subtractive primary colours. For example, cyan and magenta paints or two filters on top of each other will produce blue, since the primary green part of cyan and the primary red of magenta are subtracted out by the other filter. This leaves only primary blue going through both filters to reach our eyes. People often say that red, blue and yellow are the primary colours for paints. This is a kind of approximation but, in fact, it is incorrect! Really the primary colours for paints are magenta, cyan and yellow.

The colours of objects under different coloured lights

A banana is yellow (secondary colour) under white light because it reflects back green and red (making yellow additively) and absorbs the blue end of the spectrum. Under yellow light, it will also appear yellow. Under red light, it will appear red, as it reflects red light. But under blue light, it will appear black as it absorbs the blue light.

For similar reasons, a red (primary colour) apple will appear black under all lights except white and red. This is because red is a primary colour and only red is reflected.

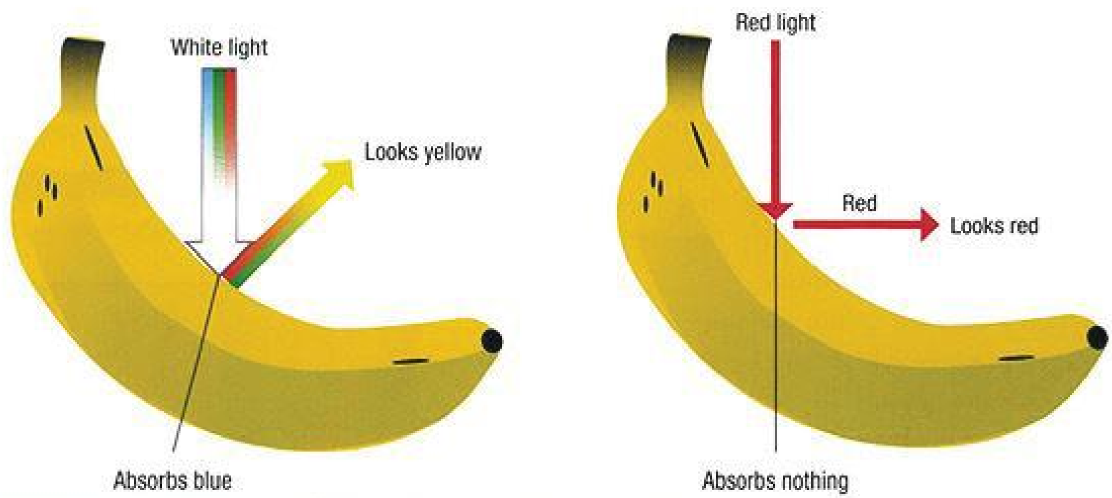

FIGURE 1.33 Why a banana has different colours under white and red lights

REVIEW

1 Which of the statements are true and which are false? Where a statement is false, rewrite it to make it correct.

- a If an object does not produce light or reflect light, it is invisible.
- b White light is really made up of many different coloured lights.
- c A magenta spotlight and a green spotlight can together produce white light.
- d A red filter stops red light from passing through.
- e Complementary colours are colours that look good together.
- f Mixing red, green and blue lights always produces white light.

>

2 Outline how well do you think our eyes are protected.
3 Figure 1.34 is a diagram of the human eye. Sketch the diagram in your book and label each of the parts indicated.
4 Outline why some objects look white while others look black.
5 Explain how our eyes enable us to see the colour yellow.
6 Grass changes colour from green during the day to black at night. Discuss. In formulating your answer, be sure to consider how the eye works.
7 a State the primary colours for additive colour mixing.
  b State the names of all the complementary colours.
  c State the names of all the subtractive primary colours.
8 Figure 1.35 shows the overlap of three coloured filters – cyan, yellow and magenta. The filters are illuminated by white light. Identify the colour seen in each of the four overlapping areas.

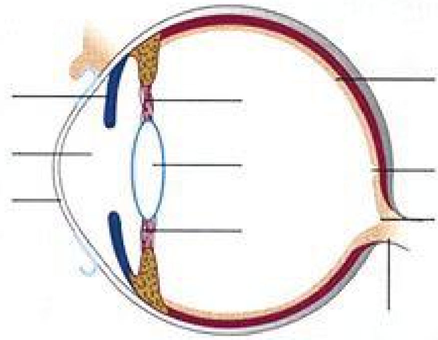

FIGURE 1.34 Diagram of a human eye

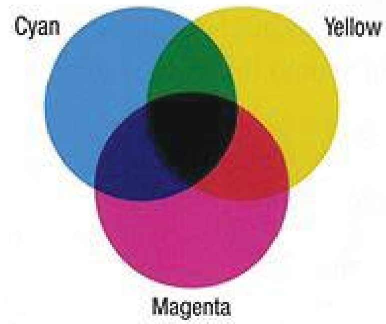

FIGURE 1.35 Secondary colours overlap in white light to produce different colours.

Vision impairment

An adverse effect on vision is called a vision impairment. There are many different forms of vision impairment.

Colour blindness

Colour blindness is a vision impairment associated with the reception and processing of one or more colours. The cone system does not work properly because cones are either defective or absent. Colour-blind people cannot detect the differences between some colours (such as red and green) or do not see some colours as brightly as other people do (Figure 1.36).

Defects in colour vision can be very mild, and the person may not be aware of the problem. Most problems with colour vision are inherited. About 8% of males and 0.5% of females are colour-blind.

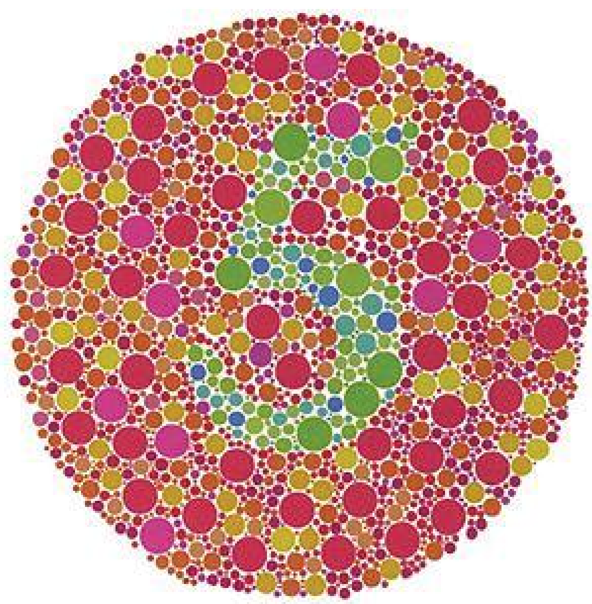

FIGURE 1.36 What do you see here? A colour-blind person will not see the number in the centre of the circle.

Vision problems and refraction

Go to http://mypphys45.nelsonnet.com.au and click on **lenses**. Try out concave and convex lenses to correct two types of vision problems.

About 145 million people worldwide suffer from refraction problems – short- or long-sightedness or astigmatism – that can be corrected with spectacles.

Short-sightedness (myopia)

Short-sightedness, or **myopia**, occurs when people can see objects clearly a short distance from them, but objects in the distance appear blurred. This is because the light from a distant object focuses in front of the retina, causing blurring. The eyeball is too long from lens to retina (Figure 1.37a). Myopia can also be caused by the inability of the ciliary muscles to cause the lens to change shape.

Myopia is corrected by placing a concave lens in front of the eye lens. The concave lens diverges the rays on their way to the lens of the eye. The eye's lens is able to focus the image further back onto the retina (Figure 1.37b).

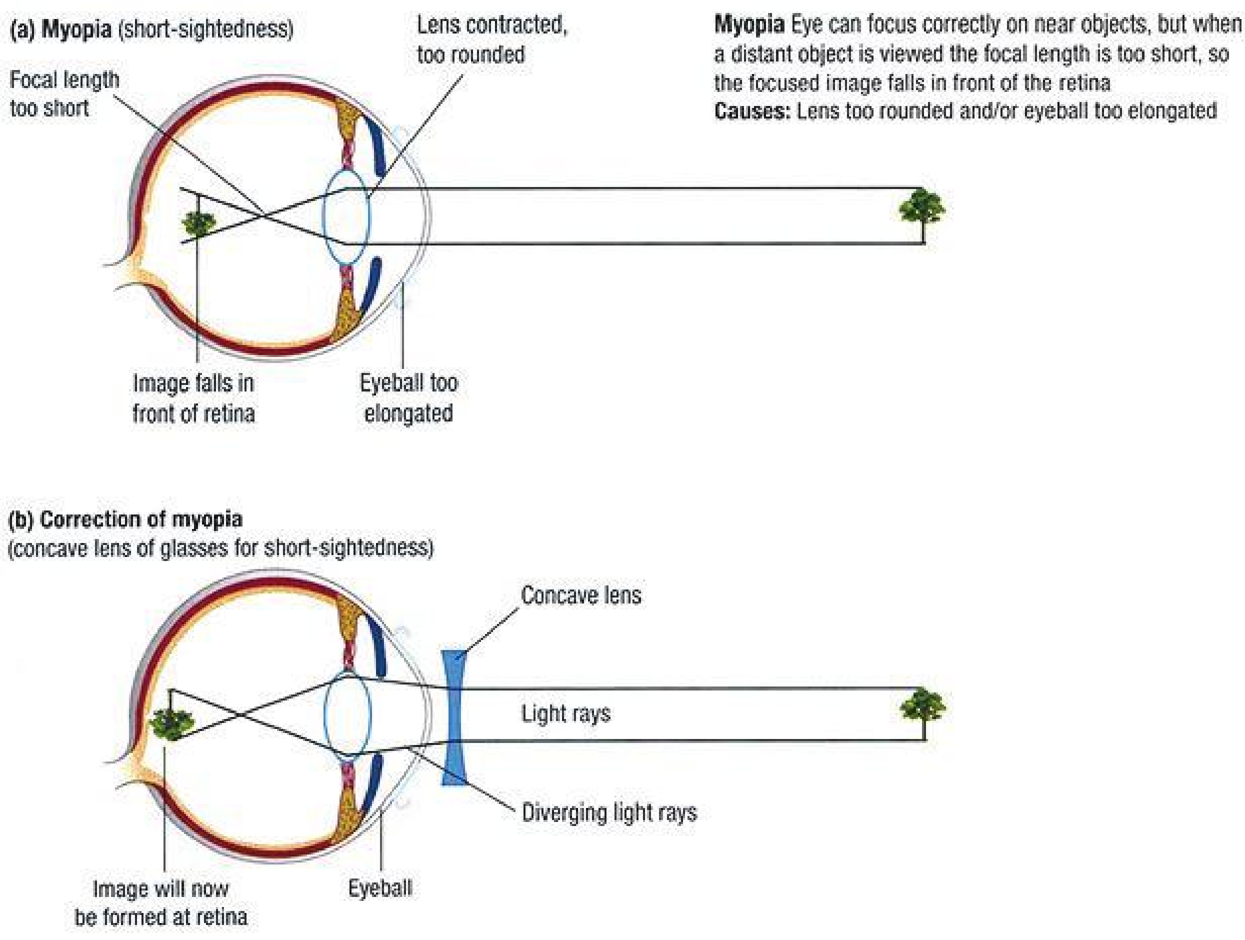

FIGURE 1.37 Myopia. (a) The image forms in front of the retina. (b) This can be corrected with a concave lens.

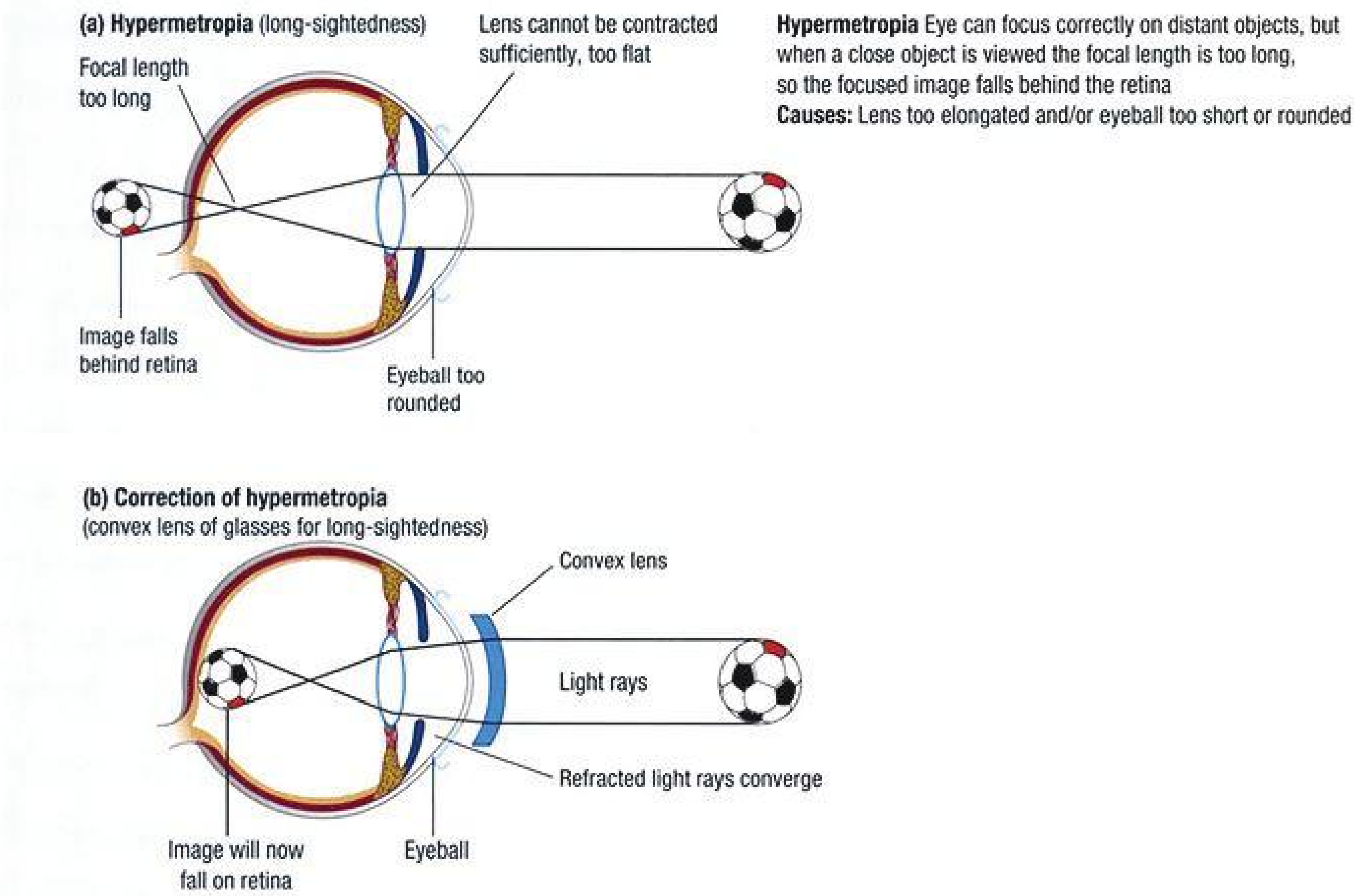

FIGURE 1.38 Hypermetropia. (a) The image is formed beyond the retina. (b) This is corrected by placing a convex lens in front of the eye.

Long-sightedness (hypermetropia)

Long-sightedness, or **hypermetropia**, occurs when people can see objects clearly a long distance away, but close objects appear blurred. This is because the light from a near object focuses towards a point behind the retina, causing blurring. The eyeball is too short from lens to retina (Figure 1.38a). Hypermetropia can also be caused by the eye's inability to accommodate.

Hypermetropia is corrected by placing a special type of convex lens in front of the eye. The convex lens converges the rays on their way to the lens of the eye. The effect is that the eye's lens is able to focus the image further towards the front of the eye onto the retina (Figure 1.38b).

Astigmatism

Astigmatism distorts the image formed in the eye because the lens is egg-shaped rather than spherical. The lens has two different focal points, one for the horizontal and one for the vertical part of the image (Figure 1.39).

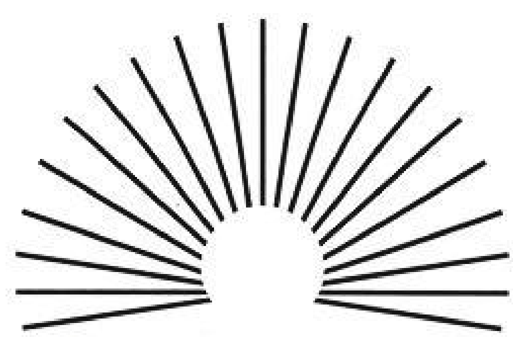

FIGURE 1.39 Astigmatic fan chart. Normal eyes see all lines properly focused. Astigmatic eyes see some lines better focused than others.

Vision problems and public health programs

Vision problems for about 70 million people could be reduced or completely cured by adequate public health programs, such as the control or eradication of eye diseases caused by **bacteria** and **parasites**. A simple operation to replace a damaged part of the eye is sometimes enough. How does our environment affect sight? What do you think is the effect on eye health of bacteria and parasites? How do you think we can ensure that eye health problems are reduced for everyone?

FIGURE 1.40 A cataract in an eye

Cataracts

A **cataract** is a cloudiness in the eye that, left untreated, can cause blindness. Look at Figure 1.40. Notice the light-coloured cloudiness in the cornea. Causes of cataracts include medical conditions such as **diabetes**, injury to the eye, medications such as steroids, various chronic eye diseases, and long-term, unprotected exposure to sunlight. There may also be genetic causes.

Generally, cataracts can be readily cured with surgery. But if cataracts are left untreated, the person will go blind. Even if they do not go blind, people with cataracts find it harder and harder to carry out their day-to-day lives.

World action on cataracts

Go to http://mypphys45 nelsonnet.com.au and click on **Fred Hollows Foundation** to find out more about the Fred Hollows Foundation and the impact of affordable ophthalmoscopes on remote villages in Africa.

A project started in 2009 by the International Centre for Eye Health aims to identify the level of vision in cataract surgery patients. This was to be used as baseline data to assist in getting rid of cataracts by the year 2020. The Fred Hollows Foundation started treating cataracts in the Indigenous population of Australia. It has expanded its work into Africa, South and South-East Asia and the Pacific. The Foundation has helped develop the Arclight, an affordable, solar powered **ophthalmoscope** that allows eye doctors to diagnose eye conditions. Cataract surgery is a fairly simple procedure that can have far-reaching effects. A cured patient no longer needs a family member to look after them. This person can return to school or work and contribute to the family's finances.

FIGURE 1.41 *Chlamydia trachomatis* is a bacterium that causes trachoma. It cannot live outside a host cell

Trachoma

The bacterium *Chlamydia trachomatis* causes one of the oldest known diseases, **trachoma**. Trachoma affects about 80 million people. It is transferred by rubbing the eyes with the fingers or dabbing the eyes with an infected cloth. It is transferred between people by little flies (*Musca sorbens*) that drink the moisture from eyes (Figure 1.41).